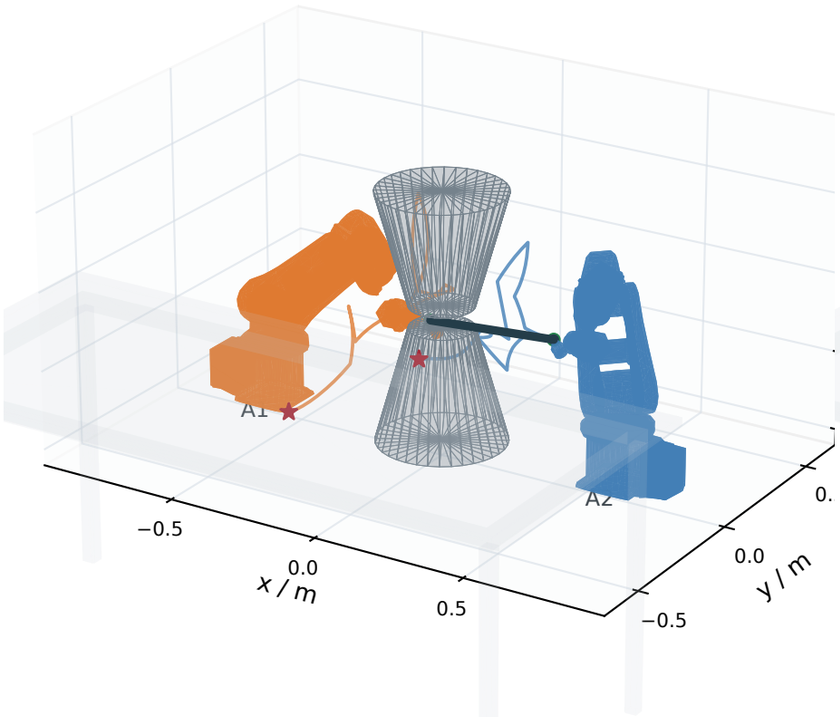
Environment III | narrow passage and four active-arm role changes

start | node 1/649

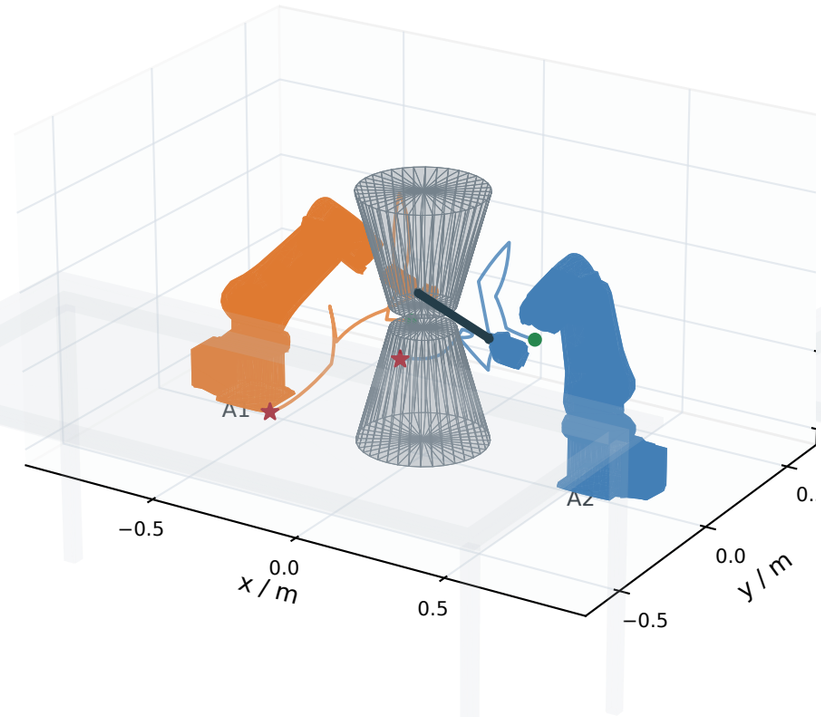
saved midpoint | node 325/649

goal | node 649/649

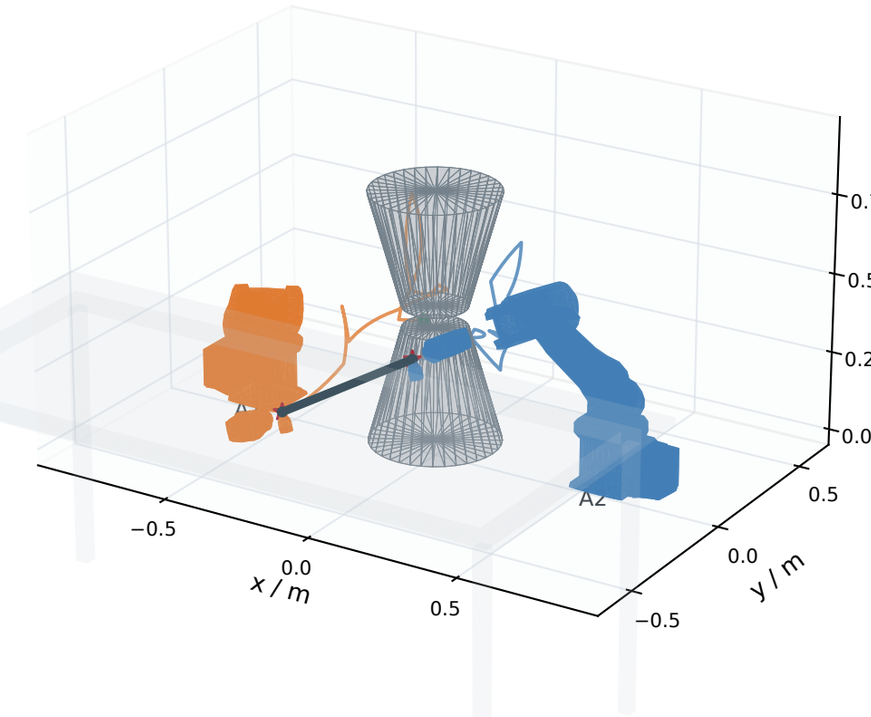
active A1; passive A2



active A1; passive A2



active A1; passive A2



Two original cone surfaces: lower (identity) and upper (Rx=3.14, z=0.790 m). Base separation 1.200 m; grasped bar 0.500 m.

Saved evidence: closure max 4.55e-13 mm; rotation matrix max 4.47e-15; 649/649 aligned TCP poses pass; 0/649 native poses pass.

The four role changes show the mechanism on this final path. A performance advantage over compatible fixed PCS or NR is not established.

Aligned reconstructed discrete source collision policy only. Obstacle transparency is for visibility; no clearance or continuous CCD certificate.